

Phq-2 And Phq-9 For Detecting Major Depression

Introduction

Depression screening questionnaires recognize patients with scores above a cutoff threshold for assessment to know whether they are depressed. The Patient Health Questionnaire–2 (PHQ-2) has been recommended as a prescreen before administering the remaining PHQ-9 items. However, previous meta-analyses on PHQ-2 accuracy have not included PHQ-2 data from additional studies nor evaluated PHQ-2 accuracy in combination with the PHQ-9 (Levis et al., 2020). This meta-analysis of single-participant data aimed to evaluate PHQ-2 screening accuracy in adults using different reference standards among participants verified as not diagnosed or in treatment alone and combined with the PHQ-9 vs. the PHQ-9 alone.

Literature Review

Major depression is a common and debilitating mental illness affecting millions worldwide. Early detection and treatment of this condition are crucial for improving patient outcomes. One of the tools used to screen for depression is the Patient Health Questionnaire (PHQ), which consists of two versions - the PHQ-2 and the PHQ-9. This literature review aims to examine the accuracy of the PHQ-2 alone and combined with the PHQ-9 for screening to detect significant depression.

Accuracy of PHQ-2:

The PHQ-2 consists of two questions about the presence of depressed mood and anhedonia over the past two weeks. Studies have shown that the PHQ-2 has a high sensitivity for detecting major depression, with reported values ranging from 75% to 100% (Mitchell et al., 2018; Thombs et

al., 2018). However, the specificity of the PHQ-2 is lower, ranging from 42% to 86% (Mitchell et al., 2018; Thombs et al., 2018). Overall, these findings suggest that the PHQ-2 is a valuable tool for identifying individuals who may be experiencing major depression, but a positive screen on the PHQ-2 alone does not confirm the diagnosis of depression.

Accuracy of PHQ-9

The PHQ-9 is a nine-item questionnaire that asks about the frequency and severity of depressive symptoms over the past two weeks. Studies have shown that the PHQ-9 has high sensitivity and specificity for detecting major depression, with reported values ranging from 79% to 94% for sensitivity and 81% to 95% for specificity (Mitchell et al., 2018; Thombs et al., 2018). The PHQ-9 has also been shown to be a reliable tool for monitoring the severity of depressive symptoms over time (Kocalevent et al., 2018).

Accuracy of PHQ-2 and PHQ-9 Combined:

Several studies have investigated the accuracy of combining the PHQ-2 and PHQ-9 for screening to detect significant depression. One study found that using a positive screen on either the PHQ-2 or the PHQ-9 had a sensitivity of 93% and a specificity of 56% (van Steenbergen-Weijenburg et al., 2018). Another study found that using a positive screen on both the PHQ-2 and PHQ-9 had a sensitivity of 69% and a specificity of 94% (Aguilar et al., 2018). Overall, these findings suggest that combining the PHQ-2 and PHQ-9 may improve the accuracy of depression screening, particularly when both instruments are used to confirm a positive screen.

Methodology

Ethical approval and informed permission were not necessary for this investigation, which entailed the analysis of anonymized data that had already been gathered. Studies were included in this meta-analysis if they met the following criteria (Levis et al., 2020): they used PHQ-2 scores or item data to calculate PHQ-2 scores; they classified prevailing major depressive disorder or episodes of major depression using DSM or ICD criteria and a validated diagnostic interview; they administered the PHQ and diagnostic interview within two weeks; they included participants aged 18 and up; they did not recruit participants exclusively from psychiatric settings; and they did not recruit participants from schools or universities.

Results

Just 100 out of 136 research met the inclusion criteria, and those studies provided individual-level data (mean [SD], age, 49 [17] years; 59% female). PHQ-2 sensitivity and specificity (95% CI) for cutoff scores of 2 or larger were 0.91 (0.88-0.94) and 0.67 (0.64-0.71), respectively, among studies that employed semi structured interviews and 0.72 (0.67-0.77) and 0.85 (0.83-0.87) for cutoff scores of 3 or greater. Semi structured discussions were far more sensitive than structured ones. There was no discernible variation in specificity between the various interview formats.

For semi structured interviews, the area under the receiver operating characteristic curve was 0.88 (0.86-0.89) (Levis et al., 2020). For completely structured interviews, it was 0.82 (0.81-0.84), and for the MINI, it was 0.87 (0.85-0.88). There were no substantial variations between the groups. Sensitivity for PHQ-2 scores of 2 or more followed by PHQ-9 scores of 10 or more

(0.82 [0.76-0.86]) was not statistically different from PHQ-9 scores of 10 or more alone (0.86 [0.80-0.90]) for semi structured interviews; specificity for the combination was marginally but considerably higher (0.87 [0.84-0.89] vs. 0.85 [0.82-0.87]). The AUC for this curve was 0.90. (0.89-0.91). It was calculated that between 56% and 58% fewer people would need to take the entire PHQ-9 if the two were used together.

Discussion

The article presents an individual participant data meta-analysis of 44 studies that used semi structured diagnostic interviews to evaluate the accuracy of the PHQ-2 and PHQ-9 for screening major depression. The study found that the sensitivity of the combination of PHQ-2 (cutoff ≥ 2) and PHQ-9 (cutoff ≥ 10) was not significantly different from the full PHQ-9 (cutoff ≥ 10) for all participants, but the specificity was minimally, yet significantly higher. The combination approach reduced the number of participants who needed to complete the full PHQ-9 by 57%. However, the PHQ-2 alone had lower sensitivity or specificity, depending on the cutoff score, compared to the PHQ-9 (Levis et al., 2020). The study also found that PHQ-2 sensitivity was highest compared with semi structured interviews and lower compared with fully structured interviews and the MINI. The specificity estimates were not significantly different across reference standards. The study suggests that misclassifying significant depression among participants with subthreshold depressive symptoms based on fully structured interviews might explain the lower sensitivity compared with semi structured interviews.

Conclusion

This meta-analysis offers a complete assessment of the screening accuracy of the PHQ-2 and lends support to using the PHQ-2 as a primary diagnostic instrument for depression. It has been found that adding PHQ-2 to PHQ-9 does not significantly improve the accuracy of depression screening; hence, using PHQ-9 by itself may be adequate for depression screening. When comparing PHQ-9 cutoff scores of 10 or more with PHQ-9 cutoff scores alone, a meta-analysis of trials with individual participant data indicated that a combination of the PHQ-2 (with a cutoff of 2) followed by the PHQ-9 (with a cutoff of 10) had identical sensitivity but greater specificity.

References

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